ActInf Textbook Group ~ Cohort 3 ~ Meeting 4 (Chapter 2, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 4 (Chapter 2, part 1) | 55:06

all right hello

it's February 15 2023 we're in cohort 3

of The Textbook group

and we're in our first discussion on

chapter two of the textbook

so

there's a lot to discuss chapter two

goes hard

before we

go into the questions that people have

raised

does anyone want to just share a first

recollection something that stuck with

them or a friction or just a sentiment

or a feeling that they had when they

were

reading even part of chapter two

you can just unmute and go for if you

want to share

it's awesome that so many people added

questions there's going to be a lot to

discuss

anyone just want to give a first

thought

what did they what did they hope

chapter two would do

was there any section or like quote or

what

stood out when people were on the low

road to active inference

okay

let us

skim

in a few minutes the whole chapter and

then we're gonna turn to the questions

and just go based upon which questions
people have
voted as interesting
but let's just take a quick uh flyby of
the low road
so chapter two
is going to be coming on the low road
so just to get a visual
reminder of what we're talking about
with the high road and the low road
from figure 1.2
the low road is like the how
showing how from Bayes theorem which is
so simple and tautological and
contentless
we get a how
that meets with a y
which is this imperative for persistence
and self-organization and they come
together in the generative models used
in active inference which we're going to
get to in chapter four so the low road
is going to talk about a lot of the how
and some of the what I guess as well
chapter three is going to come from The
High Road perspective and chapter four
is where we're going to come to the
heart of active inference with the
generative models
thank you Maria we'll get to 2.3 and and
then feel free to address that
section 2.2
is perception as inference
and this section gives a little bit of
an overview on how even before the
Bayesian brain
was described
there was concepts of perception as

inference with helmholtz and even
further back as action Maria shared in
live stream 43.0
and what active inference is doing is
extending that inferential framework
Beyond perception to also include action
that is in the title it's active
inference it's about inference on
perception and action
this is going to be operationalized with
modern statistics
throwba is called a generative model
we're going to come back through this so
just moving rapidly
probabilistic reasoning
is described by this equation
which is
um it'll be awesome I think as a
question asked like to have this in
plain language what is being described
or what is being done with basic
question
an example is introduced
where a person is holding either a frog
or an apple in their hand and then the
object is going to either jump or not
jump
the person has some prior beliefs about
a priori prior beliefs How likely frogs
and apples are in the world
and they have some beliefs about what is
likely to do what action
an observation comes in and then one is
able to have a posterior belief or a
posterior a posteriori
of How likely they think the object is
to be one thing or the other they've
updated their prior beliefs

through observation so like the Bayesian
moment is this critical moment where a
new observation comes in
and that in the context of a likelihood
model
updates the prior into a posterior
here's an exact Bayesian inference on
that
um scenario and for simple settings
exact Bayesian inference is totally fine
it turns out that for larger and more
challenging
statistical areas one has to use
approximate Bayesian inference
but what approximate Bayesian inference
approximates is exact Bayesian inference
so this is kind of like the core of what
you could do if you had infinite
computational power
and you can do it for simple examples
there's a discussion of probability
distributions because there's multiple
probability distributions as we're
dealing with probabilistic variables
people may be familiar with the gaussian
or the so-called normal distribution or
bell curve
however there are other probability
distributions
that cover a different space on the
x-axis
that's called support
and ultimately just have different
shapes
so sometimes you want to be reducing
your uncertainty about a bell curve
other times you might want to be
reducing your uncertainty about other

distribution families
like waiting times
there's a discussion of two kinds of
surprise there's kind of plain surprise
or surprisal which is just
how surprising was that observation
and it doesn't imply any updating of the
generative model or belief it's just
like taking a look at that observation
alone how surprising was it we call that
surprise surprisal and it's measured in
information theoretic units a Nat is
like a bit
but a bit is like a zero or one coin
flip a Nat is like a bit but it's like
your coin has e sides 2.7
Etc sites so that's surprise just simply
how surprising is an observation and
then there's also Bayesian surprise
Bayesian surprise also measured in Nats
is about how much
the observation updates the prior into
the posterior so they're both measured
in information theoretic units
surprisable is describing
um how surprising an observation is with
no reference to the updating and the
second notion of surprise is the
Bayesian surprise which is how much
beliefs update
expectations are described in box 2.2
although conversationally we might talk
about like are you expecting it to rain
tomorrow
and expectation of a distribution is not
a prediction about how it will be in the
future
necessarily

expectation is the just arithmetic mean
so it's just the center of gravity the
center of mass
of a probability distribution is its
expectation
having introduced a more General
Bayesian and information theoretic
perspective on
just inference it's gonna return to
biology
so Maria asked what is
um how does active inference relate with
Biology and psychology was that the
point of section 2.3 so does anybody
want to give a thought like what was
section 2.3 doing or how is
optimality and Bayesian statistics
related to biological inference
either restating what they added or just
adding your own angle on it
Ali go for it
uh I think section 2.3 uh serves as a
kind of interlude between 2.2 namely
perception as action and 2.4 action uh
sorry perception as inference and 2.4
action as inference because uh before
dealing with the concept of action as
inference we need to be clear uh what
kind of inference we're dealing with
here namely uh it's not just uh any kind
of uh inference that's for example been
discussed in other various uh
literatures and statistics or in other
areas but we're dealing with some
specific kinds of Bayesian inference uh
which uh is uh one of its main
characteristic is being optimal because
uh to be optimal

in this framework is a necessary
condition for
um for and for inference because uh
without optimality we wouldn't have any
um
uh actionable
um state of knowledge uh so
um that's one of the main points of 2.3
uh because uh I believe the concept of
optimality is a necessary requirement
for any biological or even
non-biological agent to act as a
sentient agent in any kind of uh
environment so that optimality is a kind
of
at least in this section it's been
described as a kind of requirement for
any kind of sentence sentient Behavior
thank you awesome
great way to put it as like an interlude
because biological or even the kinds of
active systems that aren't necessarily
biological we want to talk about
inference on perception but also we want
to talk about inference on action
and part of the unifying capacity of
active inference is that perception and
action are going to be in service of a
common imperative which is free energy
it's a really important note that
optimality
does not imply that it's adaptive it
doesn't mean the organism succeeds it
doesn't mean it even works so it's
really worth it to see what is meant by
optimality
optimality is defined in relation to a
cost function that's optimized

that's the variational free energy which
is going to be related to surprise which
is why surprise was brought up
as important earlier
so it's also unpacked a lot more in the
neuropsychiatry context
where instead of saying well healthy or
neurotypical people have optimal
inference and then it's sub-optimal
inference for those who are different in
this or that way
rather in the Precision neuropsychiatry
setting
we can talk about Bayesian inference
being optimal
however with different generative models
the outcomes of optimal inference are
going to look different
if somebody has an 100 belief that it
rains every day
then their optimal inference will lead
to them believing that it rains every
day so this doesn't mean Society accepts
doesn't mean the business succeeds
doesn't mean that the model works better
than any other model it is just a
statement about how Bayesian inference
is going to unfold and as Ali pointed to
this is helping us
um move from a perceptual
inference model into a subjective
subject-specific generative model of
perception and action which can be
described as sentient in the sense used
by fristin at all
to mean adaptive and responsive to
stimuli
figure 2.2 is going to introduce a

really important distinction between the
generative process and the generative
model so the generative model is the
cognitive agent
it's there whether you think of it as
they are or they have a generative model
that's like a realist angle or we're
making a generative model for a
cognitive agent that's the
instrumentalist more map territory
falling more on the map side so
generative model is a Bayesian
statistical model
that is being used to describe
a cognitive agent
the generative process
can be thought of as the niche that is
the underlying process passing
observations to the generative model
the niche the generative process can be
an active inference process itself but
it doesn't have to be
so the generative process could just be
an if then
it could be any kind of logic you can
have any endogenous Dynamics
however it passes observations
through the Markov blanket as sent
states to the generative model the GM
section 2.4
the discussion to this point is common
to all Bayesian brain theories
we now introduce the simple but
fundamental Advance offered by active
inference alarm bells ringing
it considers action as inference active
inference is like
Bayesian brain

plus
pragmatic turn
Bayesian inference perspective
on action
that gets unpacked a little bit more
in the next two sections we clarify the
single quantity that active inference
agents minimize through perception and
action is variational free energy
so whereas a reward learning
cognitive agents
would be maximizing reward
and active inference agent as modeled as
the way we're going to be exploring in
this textbook
they have two ways
to approach a single imperative
the single imperative is variational
free energy
the two ways that variational free
energy can be minimized
is changing your mind and changing the
world
perception and learning and action
so first
just minimizing the discrepancy between
the model and the world
at a first approximation the common
objective of perception and action
can be formulated as a minimization of
the discrepancy between the model and
the world
sometimes this is operationalized in
terms of prediction error
so here's a
more classical prediction error
minimization this is also reminiscent of
like t-o-t-e

Cycles looking for deviations and then
acting to resolve deviations or
discrepancies
between predictions of the generative
model observations coming in from the
generative process and then here are the
two ways that discrepancies can be
reduced
perception change belief oh I didn't
think it was raining but it's wet
discrepancy alarm Bells surprise
surprisal
surprising observations
so then what can be done well one can
change belief
so that upon the next cycle
there is no discrepancy
or there are situations where action can
be taken so that future observations are
in line with the unchanged predictions
so
those two pathways
perception and action
are going to come back again and again
inactive inference
how much should we take the idea of
cycle seriously from Jonathan presumably
it's much more non-linear
yeah that's an awesome question does
anyone want to
give a first thought on that
there's there's a bunch of ways to
explore this
one is although sometimes we see it as a
circle this is actually a very informal
representation
two representations or ways of thinking
about it that are a bit closer to

formality
one
we're going to come to in figure 4.3 and
chapter 7 as well which is when you
implement this computationally it's
going to be like unrolled as a linear
sequence
so one answer is computationally you're
going to unroll the ways that these
variables update each other
and they still May describe non-linear
functions so we still can get a
non-linear function being described at
in various places even if it's unrolled
and then the second formal
representation
that helps us escape this sort of like
loops
situation is
there's going to be
um not as much in this textbook but in a
lot of the recent work on Bayesian
mechanics
equations are defined
for these variables circles random
variables as flows on those
partitioned variables
so in that setting
you can draw the edges about how
different variables flows influence each
other but it's actually like an
unfolding flow that just continuous
across four different partitions
that's the particular partition
and so you don't need to draw any kind
of like looping
but the action perception Loop is
commonly

understood
and it's also very deep topic
Ali go for it
uh sorry yeah I think uh for this
question uh that figure from
um path integrals strange kinds paper
might also help to uh illuminate this uh
concept of cycles and uh the relevance
of this concept uh to different types of
particles uh that have been described in
this paper so yeah I just wanted to
point it out to that diagram yes here's
the flows on different states
and then the diagram
looking a little inverted but continue
what would you like to say about it
so basically uh in this paper they have
defined four different types of
particles the simplest one being inert
particle
which doesn't have any active States and
as we increase uh in terms of
um I mean complexity or uh increasingly
more sentience Behavior we're dealing
with active particles which have uh
active States but uh
as opposed to strange particles which do
possess the
disposition to being actually a sentient
particles in case of active particles
and conservative conservative particles
there isn't any hidden active States so
that's basically uh
encompasses uh the different
self-organizing or even non-self
organizing uh entities ranging from
simple rocks to brains and organisms so
I think that's a pretty good way to

somehow uh
a frame frame all of those behaviors
within that Bayesian mechanical
framework
thank you ever
I have a question of uh the figure 2.3
um it looks as if there's always
discrepancy but I can also imagine that
I have a perception of objects in the
world say for example where there isn't
any discrepancy between my prediction
and my sensory observation but that
doesn't is not shown by this picture or
is it
the discrepancy value could be zero and
especially for in silico simple examples
like where the generative model is the
exact same structure as the generative
process
then the discrepancy can be zero I
predicted a coin flip heads it came up
heads no discrepancy I don't need to
change my belief maybe I'll strengthen
my belief and I don't need to take any
further action so discrepancy value can
be zero however
in the context of fitting a generative
model
that is a coarse grains or partial
representation
of
potentially noisy data coming in there's
always going to be
a non-zero discrepancy
so that's why we're interested in
relative discrepancy reduction
but there's also contrived examples
where the discrepancy could be down to

zero
but there are situations where whether
the discrepancy is zero or low or
already as minimal as it can be that
there may be no perception change there
may also be no action change or you
could imagine observations coming in
that induce large changes in belief High
Bayesian surprise and
are associated with the agent taking
action
thank you earlier in the case of the
object jumping out of the hand or not
there was equations showing exact
Bayesian inference plug and chug just
push the numbers through the Bayesian
equations
yet exact Bayesian inference supporting
perception action is computationally
intractable in most cases and they're
going to describe the cases more
it can just be understood coarsely as
being hard to know the total State space
of what could happen
and it might require taking an integral
or sums over functions that are very
like large or intractable so
it can be challenging to do exact
Bayesian inference and so one of the key
moves in active inference as most
currently use it today
is variational inference
it will be unpacked in chapter four when
we talk more about the generative models
it suffices to say that performing
variational Bayesian inference implies
substituting the two intractable
quantities

posterior probability and log model
evidence
so that's like what we would want to
know if we had full access to
information infinite computational
resources
doable for simple problems intractable
for higher dimensional State spaces and
so on
we're going to replace those two
posterior probability is going to be
replaced with an approximate posterior
called Q
and then log model evidence is going to
be substituted functionally by a free
energy calculation this is called
energy-based learning broadly
it's been around for decades active
inference does not invent the idea of
using a free energy
value
on a Bayesian statistical learning or
inference challenge
what is different is inference on action
but it is not something that was
generated within the octave inference
Community it's existed for decades
variational free energy variational Bays
is used broadly and every day
octave inference uses inference on
action and perception
that's what integrates that kind of
perceptual signal processing elements
with the action selection control
theoretic element
and
it uses the particular partition
which is the partition of the cognitive

agent from the niche
to operationalize variational inference
in this kind of agent-based
perception cognition action Loop
or just scenario so it's always
important as we're like learning through
and again multiple coats of paint but
what is actually new and different in
active inference
what is describing or condensing or
distilling work that is totally
non-contentious and has existed for
decades and is being used day to day
evidence lower bound and energy-based
learning people will find copious
examples of
so that's one deflationary take to use a
fristianism
on active inference is doing variational
Bayesian inference isn't novel
unified imperatives for perception and
action
are also not necessarily novel
but using variational inference
on the particular partition
so that there can be a unified
imperative for perception cognition
action
is when we actually get to active
inference
equation 2.5
describes variational free energy we're
going to come back to that
figure 2.4 gives a visual representation
of how variational free energy bounds
surprise
so here's like what we would want to
know

here's how we can approximate or bound
what we want to know
figure 2.5 revisits that discrepancy yes
ever at first
yes I I have some difficulty with this
founding uh can you explain to me
yes this figure uh what is meant by PMG
bounds
what what about
awesome so I'll just read what someone
else has already added and then anyone
else feel free to like raise your hand
to add more so
um there's something we can't actually
calculate but we can calculate something
approximately
we know that whatever value we get for
the approximate thing
the True Value we're looking for
is certainly lower than the thing we can
calculate so we want to know if the
organism's age
you don't have any information about
them but you do know the organism can
never live to be over 10 years old so
that's an upper bound on what you
believe the age of the organism to be
so surprisal so Ln is the natural log so
natural log and you can undo and do
natural logs with exponentiation
not going to go into too much detail on
that but it's just an operation like
whether when you see something Ln and
then parentheses you can
um take it in and out we're going to
talk more about logs later
um so this is what we would really want
to know

how surprised should we be
for a given observation
because if we knew how truly surprised
to be for any given observation
we would have access to that true
underlying distribution
that is controlling how surprised we
should be
like if you don't know the the range in
temperature
you don't know how surprised to be for
different observations if you knew the
underlying temperature distribution you
would know exactly how surprised to be
with different thermometer readings
what energy-based learning does again
this is not an active inferencesm this
is energy-based learning or evidence
lower bound
used broadly in Bayesian statistics is
it constructs an approximatable
functional
which is a function of a function
an approximatable functional free energy
variational free energy equation 2.5
that's a function of r approximation
which is a function that's why f is a
functional and data
so instead of trying to find the
absolute
underlying surprise how surprised we
should be by a data point which would
entail knowing the true distribution
we're going to construct this functional
variational free energy that's a
function of
q r approximation and Y the data
and then this is a KL Divergence

the double line
what is on both of the sides the double
line is what is being Divergence
minimized
we're trying to minimize the Divergence
between
our approximation
and the true distribution conditioned on
data
this is guaranteed to be an upper bound
it's always higher than the surprise
and then by doing diversions
minimization
with KL Divergence
it's possible to
push that upper bound lower
so it's like here's how much the bank
account actually has okay I know it's
less than a billion because that's
there's not that many of this token in
the world
is it less than a thousand is it less
than five so it's like as you push the
upper bound down
you're able this can be approximated
well again whether your startup works or
whether the organism lives or whether
the model's adaptive or whether it's
better than any other model anyone else
could ever think of that's an it's not
this question
this is saying you can generate
a derivative not in terms of like a rate
of change but like a sort of corollary
value
based around data and our approximation
that's always higher than surprise
so surprise was brought in early in the

chapter
as a key imperative to surprise minimize
but then that's intractable
so we have an approximatable tractable
form
and that's what energy-based learning is
and it's not invented in active
inference
but what is a little novel and different
is doing energy-based learning
Bayesian variational inference
on action
we revisit in 2.5 figure
free energy
and instead of each being discrepancy
here which is kind of like an a
prediction error minimization that's
like the tote
like is there discrepancy reduce the
discrepancy well discrepancy between the
predictions and the observations when
you think discrepancy you're thinking
like it's going to be like I predicted a
seven I got a nine the difference was
two
that would be prediction error
free energy is doing energy-based
learning and so it's like a discrepancy
because if the data coming in
were exactly what was predicted
this Divergence would be zero and this
term would basically
kind of functionally do what you would
think a prediction error would do in the
situation where the prediction matches
the observation
but it turns out to also extend in a few
other ways but the reason why they

showed the discrepancy first in 2.3
and then you can change perception or
change your auction selection
and then
prediction observation come together
calculate a discrepancy and then
decisions are made
cognitive decisions
perception or like embodied decisions in
action
that change the actual generative
process
and then in 2.5
that gets built out a little bit beyond
just prediction error type discrepancies
like there's a four inch discrepancy
between these two things into an
information theoretic or a statistical
quantity
that's variational free energy
f
section 2.7 talks about expected free
energy and planning as inference
you can't talk about variational free
energy on data points that haven't come
in yet because their future observations
and so expected free energy G
is similar
and it's explored in 2.8 it's similar in
certain ways
but importantly and like we return to
this chapter after chapter so no worries
how much
of these equations
look
fluent to you right now
one thing is these equations have
natural language representations that

people can also improve but especially
like 2.5
for f
you can read it and see the active
inference ontology terms being used
rather than variables and then in 2.6
it's um some part of it is written but
like these are the kinds of
augmentations that are always very
welcome for people to to make because
they help us understand
um expected free energy is like
variational free energy but whereas
variational free energy is real time and
it's crunching our approximation with
the incoming data
expected free energy is calculated over
policies which are sequences of action
and it is going to be dealing with
observations that haven't yet come in
so this is where we get to policy
selection
in active inference by thinking about
expected free energy which is like a
prospective
talking about the expected free energy
for different possible policies which
are sequences of actions
there's some more discussion about
special cases
expected free energy
when certain terms
are basically zeroed out
expected free energy becomes resembling
other equations so this is like a common
technique in math
you have some equation and then like you
can generalize it and then a special

case of the general form is this and
then another special case might be that
so for example
we'll come back next week or in the
future
to what exactly the pieces are um
representing
but one can imagine that if pragmatic
value
was irrelevant
then
G would reflect Information Gain of
different policies
if information gained were irrelevant
like it were a totally discovered
situation
then G would reflect purely pragmatic
value
so G is like a generalization it's a
unified imperative
under which certain special cases
resemble variously
information maximizing principles
or utility maximizing principles
Jonathan is there any equivalent to
discounting as in an RL context
do you want to unpack what you mean
there by discounting but then yes
there's definitely some angles on that
yeah sure so um in a reinforcement
learning context
um you know there's there's a clear aim
which is to maximize the expected future
returns
um but generally that's done in a
discounted way and so the agent is
trying to maximize the discounted
expected future returns such that

rewards that it gets in the future are
essentially worth less than rewards that
it gets in the near term so I know that
there isn't the same kind of idea of
rewards as there is in in reinforcement
learning and active inference context
but I'm interested to know whether there
is any sort of idea of discounting that
states further into the future or
somehow less important than states in
the moment
yes here's a
um I'll I'll add that um
uh question
and then add in the um one
Briston
paper
it turns out that yes it's possible
to to get temporally discounted type
Behavior so like a simple way to think
about that would be if the
um pragmatic Drive were urgent then you
would see small pragmatic gain being
approached
at the loss of future gain
short time Horizon models and Urgent
desire Drive
whereas under longer time Horizons
longer time preferences
then you see the kind of behavior that
in utility driven decision-making is
associated with what is called temporal
discounting
so the phenomena the cognitive
behavioral phenomena
exists and and can be recapitulated
but it is not generated with the same
way that

it is
in other places
sure thank you
2.9
is the end of the low road
variational free energy
real time unfolding F of Q and Y
variational free energy our approximate
posterior q q that's the one we control
Y data just like a regression y equals
MX plus b
q and Y discrepancy except it's a
variational free energy it's real time
expected free energy G of Pi
G is a function of policies
and we're going to go into that more
but it is regarding observations that
haven't come in yet
so it has a different structure but some
rhyming with variational free energy
what do they achieve together
where did we get at the end of the low
road
so
we started with Bayes theorem
generative model could be the the thing
jumping out of the hand
perception is inference
that goes back to helmholtz but as
livestream 43 unpacks this has been
understood broadly
across the world for thousands of years
predictive coding
helps operationalize perception as
inference because it's like top-down
predictions bottom-up sensory
observation information that's why we
don't perceive our blind spot

to bring in action
we have to talk about perception as
inference because we do care about
observations
but we have to talk about observations
that haven't actually happened yet and
we have to talk about how our actions
influence
which observations are likely to occur
perception and action as inference
together make up the Bayesian brain
which is like some people it means the
brain does Bayesian computation for
that's a realist take the
instrumentalist take would be we use
Bayesian statistics to study the brain
and
where exact Bays is intractable
we can use variational base
energy-based learning
that's the end point of the low road
into active inference
summary
active inference is a theory of how
living artifacts underwrite their
existence by minimizing surprise
that's what we would truly want to know
surprise if you knew how surprised to be
you would have had to have known
the actual underlying distribution
so we usually can't do that so we have
attractable proxy to surprise
variational free energy
so just like we'd want to reduce our
surprise because that would reflect
having the most optimally trained
but not over fit
belief

we want to minimize a proxy
that we know is an upper bound so by
minimizing the upper bounds
we're getting closer and closer to what
we really want to approximate
the two ways to do that are perception
and action
they motivated this
from a very first principles perspective
and then extended what is often left
only in the realm of perception into the
domain of action
more discussion
surprise
is a construct of information Theory
it's not necessarily equivalent to folk
psychology
and this comes up again and again we see
it in the ontology for like half of the
words
these were words
that are used broadly
perception action
belief surprise
and sometimes they're perfectly
overlapping
with formal definitions other times
there's a little bit of a Divergence
and they close
perception action minimize variational
free energy and complementary waste
that's real time
active inference also minimizes expected
free energy
prospectively
this completes our journey along the low
road
in chapter 3 we travel the high road

which reaches the same conclusion
on the basis of first principles and
self-organization
so we started from we have a Bayesian
model
We Didn't Start From organisms need to
exist
but it turns out by connecting surprisal
to
self-organization or really just
persistent repeated measurements
we're going to get to the same place
all right in our
five remaining minutes just next week
we're going to return
to the to the questions but I hope that
it's useful to also just walk through it
because definitely it takes like
multiple
readings and interrogatings
and there's things that like you'd want
to know in chapter two
that are going to come up later or come
up elsewhere in other papers there's
just no
single
Road or path let alone one that would be
appropriate for all Learners and their
preferences
so hopefully it was useful
to like look over it and see it that way
next week we're going to return a lot to
the questions
so
if anybody wants to do
addition of questions it's always great
you can jump into the answers and
discourse as I see many do

and add things you can upvote what is
interesting
you can enrich what people have written
by just replacing where they use a plain
text
with something that you know could be
tagged just like that
so there's all kinds of ways that like
wherever chapter two is sitting with you
now
either by generating new questions and
discourse you can add notes on chapter
two
um here
and then also in the math group which we
haven't been Super Active on we've done
some laying out of just the mathematical
skeleton
of chapter two as an example of how that
could be done so but in our closing
minutes just we can have any closing
thoughts
okay just all anyone can raise their
hand but um so in in relation to the
temporal discounting and reinforcement
learning Ali also wrote
reinforcement learning models opposed to
active are not applicable to perceptual
tasks that do not involve Rewards or
show significantly variable Behavior so
there's there's a lot there
in situations where pragmatic value
isn't a key imperative
even if that's just a sub-task like
isocating is driven by Information Gain
yes our eyes dwell on what might be seen
as rewarding
but it's Information Gain that is the

imperative for the isocating
and so
where pragmatic value or reward or
utility functions are either
non-existent or or flat
reinforcement has a hard time
and that leads to all these contrived
reinforcement bonuses and novelties oh
maybe we'll pay them for this and we'll
do that
um and significantly
um
and um learning that reward function is
very difficult
and learning that reward function
especially in like a partially
observable or dynamic or volatile
environments ultimately you need to
reward the learning anyway
so then you end up coercing what we
would just call epistemic Value
into a reward learning framework
so why not just have a generalization of
reward learning
where in fully observed situations
pragmatic value can be unabashedly
pursued
but in partially observable environments
or where the generative model
is a simplified or coarse graining
structure relative generative process
why not just authentically pursue the
epistemic value
which is what you can do with g
but can only be done in a contrived or
an ad hoc way
if you have reward as your unified
imperative

that's not to say reinforcement learning
models are not effective that again that
the agent can't play chess or that the
startup wouldn't work
it's just to be clear that the the
adequacy of a model in the world
is an empirical question
and especially in um a textbook the
points are often being made in principle
great well again
thanks for all these
questions
I hope that we can
um rejoin next week
with some thoughts and ends
resources and other questions added in
there's just going to be a lot of fun
ways to take these different questions
and we'll come back to them so
thank you everyone and if you want
um in the following hour you can come to
the Discord and we're just going to
continue just talking and chilling about
education so thank you and farewell